

Abstract

Most studies focus on the impact of new public infrastructure on local housing markets. In this paper, we avoid endogeneity issues coming from the placement of new infrastructure by studying the maintenance of existing infrastructure and design a quasi-experiment to study local referendums conducted to renew property taxes levied for road maintenance. We compare housing sale prices between similar areas that narrowly pass or fail renewal road tax levies and use satellite images to fine-tune an Artificial Intelligence (AI) vision model for classifying road quality. Our results show that local jurisdictions with close elections that decide to cut road taxes face an average loss of 11% in road maintenance funds, experience a 23% decline in road quality, and suffer a 9% drop in house prices over the course of 10 years relative to areas that renew funding. Heterogeneity analysis reveals a stronger percentage decline in urban areas and for expensive houses relative to rural areas and cheaper houses, along with evidence for dosage-response for larger tax cuts. We rationalize these findings using a Dynamic General Equilibrium (DGE) framework and show that the short-run gains from tax cuts are outweighed by the long-run capitalization of infrastructure decay, revealing how public disinvestment negatively affects property values.

1 Introduction

Roads are an important form of infrastructure investment that affect households’ amenity values, firms’ production functions, and people’s commuting costs. A great deal of existing research has focused on the effects of new roads¹, especially in developing nations, providing valuable policy insights and spurring development initiatives like China’s Belt and Road Initiative (Huang, 2016) and India’s Pradhan Mantri Gram Sadak Yojana (Asher and Novosad, 2020). However, the literature is constrained by two fundamental challenges. The first is endogenous selection as infrastructure investments are not exogenous events but are deliberate policy choices. New road placement is selected by policymakers and is endogenous to anticipated economic growth, while repavement schedules and decisions are systematically correlated with confounding local factors such as political cycles or unobserved community characteristics that independently drive economic outcomes (Duranton and Turner, 2011; Gonzalez-Navarro et al., 2023). This identification problem is exacerbated by the second challenge: measurement. Most studies rely on road quality metrics that are inherently subjective, labor-intensive, and generated at low frequency (Attoh-Okine and Adarkwa, 2013). Such data are not only prone to measurement error that produces biased estimates, but their lack of scalability limits the geographic and temporal scope of empirical analysis (List, 2022).

In this paper, we avoid endogeneity issues by studying property taxes that fund the maintenance of existing roads and by utilizing a quasi-experimental setting that arises from analyzing close elections. We establish a novel dataset that allows us to study how changes in local taxes affect local infrastructure maintenance and neighborhood house prices. We match voting data on local road maintenance taxes from the Ohio Secretary of State with financial records of cities from Ohio Auditor of State and property sale prices from CoreLogic.

¹See Ghani, Goswami and Kerr (2016), Beenstock, Feldman and Felsenstein (2016), Levkovich, Rouwendal and Van Marwijk (2016), Hoogendoorn et al. (2019), Theisen and Emblem (2021), Gibbons et al. (2019), Fretz, Parchet and Robert-Nicoud (2022). Some papers have also observed adverse effects of new roads such as state-led repression (González et al., 2025) and environmental degradation (Asher, Garg and Novosad, 2020).

Our data also includes remote sensing data of local roads in Ohio from Google Earth. Using these rich data, we first estimate the expected loss in the road maintenance budget for a local government after a tax cut from a failed renewal of road maintenance levy. We fine-tune an Artificial Intelligence (AI) vision model using more than 53,000 satellite images from [Brewer et al. \(2021\)](#) to predict changes in road quality after a road maintenance tax cut. Finally, we use the Regression Discontinuity (RD) estimator to estimate the effect of cutting local road maintenance taxes on housing prices.

Our empirical findings reveal that when a city cuts its renewal road maintenance taxes, it faces an 11% loss in maintenance funds, its road quality declines by 23% and its house prices decrease by around \$15,350 (9%). After establishing the main results, we explore heterogeneity in the effects. We find that the effects on house prices are driven by urban areas rather than rural areas, with urban areas experiencing a 7.6% decline in house prices while rural areas experience inconsistent changes in house prices. Moreover, we find an effect on the intensive margin with evidence for dosage-response based on the size of the levy. We also observe that higher-priced homes experience larger reductions in value relative to lower-priced homes. This suggests that wealthier households are more sensitive to road quality, as a poor road in front of a high-priced home is more likely to be noticed than a poor road in front of a low-priced home. We rationalize our main findings using a Dynamic General Equilibrium (DGE) framework and show how public disinvestment negatively affects property values, highlighting the trade-off between short-term tax savings and long-term private capital losses. Following [Hendren and Sprung-Keyser \(2020\)](#), we compute a conservative estimate of the Marginal Value of Public Funds (MVPF) using our empirical estimates and find a 1.4 net welfare loss to households for every dollar of revenue loss to the government. Our focus on road maintenance instead of new development, our approach of leveraging close elections as a quasi-experiment to study the effects of local road maintenance tax cuts, and our use of an AI vision model to measure changes in road quality distinguish our study from the existing literature.

Our results are consistent with other work on the impact of roads on house prices. For instance, [Gonzalez-Navarro and Quintana-Domeque \(2016\)](#) finds that paving roads in Acayucan, Mexico, increases property values by 28%, while [Theisen and Emblem \(2021\)](#) finds 13% higher housing prices in towns nearest to a new highway in Norway. Unlike [Beenstock, Feldman and Felsenstein \(2016\)](#) and [Diao, Leonard and Sing \(2017\)](#), we do not find any anticipation effects but we do find statistically significant effects starting in the fourth year after the vote and continuing at least through the ninth year. This delayed effect highlights how the consequences of road tax cuts are not immediate but gradually accumulate as it takes time for roads to deteriorate and for the decline in road quality to be capitalized into house prices.

Roadmap. The rest of the paper is organized as follows. Section 2 reviews the related literature and positions our contribution. Section 3 provides background information, establishes the road quality metrics and presents the variables used in the study. Section 4 outlines the empirical strategy. Section 5 presents the results of the study and shares the relevant robustness checks. Section 6 examines the mechanisms underlying our findings². Section 7 implements our MVPF calculation to assess the impact of this tax cut policy. Section 8 concludes.

²In this section, we highlight the main findings from the DGE framework and relegate the model details to Appendix D.

Contribution. Our paper contributes to the literature on three main fronts. First, we contribute to the literature on the effects of public infrastructure on house prices. Our quasi-experimental design complements recent evidence from developing nations, such as Mexico (Gonzalez-Navarro and Quintana-Domeque, 2016) and Indonesia (Gertler et al., 2024). By exploiting close elections for renewal referendums earmarked to fund road maintenance, we mitigate the endogeneity concerns common in the literature and estimate the capitalization of road maintenance into property values. Second, we compute a novel measure of road quality using a fine-tuned vision model trained on satellite images. Current literature has relied on measures based on Pavement Condition Index (PCI) or International Roughness Index (IRI) to measure road quality, but these are not available for many localities. Similarly, Currier, Glaeser and Kreindler (2023) measures road roughness using vertical acceleration data from a smartphone app, but this also requires large-scale data available mainly for large metropolitan areas and is not available for smaller towns. Our approach allows us to measure road quality at the local level, bypassing the limitations of existing measures. Third, by assessing the long-term capitalization of road maintenance, we uncover a dynamic trade-off between immediate tax savings and deferred infrastructure quality, highlighting the role of local government investment in public goods for sustaining property values (Oates, 1972; Rioja, 2003). While *immediate* tax cuts may initially increase house prices by lowering ownership costs and making properties more attractive to buyers, our results suggest that these savings are eventually eclipsed by the *future* decline in local public amenities.

6 Mechanisms

A key question surrounding our findings is *why* cutting road taxes would lead to a sustained decline in local house prices. In this section, we argue that these results are consistent with classical insights from the literature on local public finance, especially the work of [Oates \(1972\)](#) and [Edel and Sclar \(1974\)](#). We develop a theoretical framework using Dynamic General Equilibrium (DGE) model to explain the mechanisms behind our empirical findings. The model captures the dynamic interplay between local government budget constraints, road quality, and housing prices. Specifically, cutting road-maintenance tax levies reduces local government funds for maintaining infrastructure, which in turn constrains the government’s ability to provide road-upkeep services. These lower levels of maintenance funding result in a deterioration of road quality, which then directly and indirectly lowers housing values. For brevity, we summarize the key insights here and relegate the full model description to Appendix D. Below, we outline the three-step channel through which road tax cuts lead to lower house prices.

1. Road Tax Cuts and Reduced Road Maintenance Funds. Local governments in Ohio, like those in many other U.S. states, rely on property taxes and dedicated road levies to fund local public goods. The [Oates \(1972\)](#) decentralization theorem highlights that local authorities can generally provide public services in a manner better aligned with resident preferences than would a higher-level government, provided they have adequate revenue sources. When a renewal levy fails at the ballot box, one of these critical revenue sources vanishes. Consistent with the premise that local road quality is a local public good, losing a dedicated revenue stream substantially weakens a city’s ability to maintain or upgrade roads. Empirically, we infer the size of the drop in maintenance funds when a maintenance levy fails to renew in Table 1.

2. Decline in Road Quality. Once the road tax levy is not renewed and the budget is cut, the local government has fewer resources to maintain roads, but does it reflect in the quality of roads? We study this question and explain our method to measure road quality

in Section 3.4. Results from our AI model-based road-quality measures in Section 5.1 show a marked decrease in road quality following the cut of renewal road tax levies. The resulting deterioration in road quality is primarily seen through remote sensing data, but may be experienced via ride discomfort, declining aesthetics, and reduced usability of neighborhood roads. This is precisely what a local public goods framework would predict: fewer financial resources for upkeep reduces the quality of a local public good. Residents observe these changes slowly over time, often after four to five years, when road conditions become visibly poor.

3. Capitalization into House Prices. Falling road quality imposes a disamenity on local residents via reduced local infrastructure that is capitalized into house values (Oates, 1969). A couple of mechanisms help explain the ultimate discount in property values:

- **Appearance and Neighborhood Appeal.** Rough roads, potholes, and poorly patched surfaces reduce aesthetic appeal of the houses in a neighborhood. Potential buyers, seeing visible signs of neglect, bid less, thereby reducing the sale price of houses.
- **Travel Costs.** Lower quality roads and chronic maintenance issues translate into slower commute times for residents, particularly in urban areas. This is a direct disutility for homeowners, as time lost in traffic takes time away from leisure or work.

Short-Run versus Long-Run Trade-Off. Importantly, a trade-off emerges over time. In the short run, residents in cities that voted against a road tax levy experience immediate relief in terms of lower property tax bills²². Their out-of-pocket expenses decline, which can be a tangible financial benefit, particularly in communities where property tax burdens were already viewed as high. However, as road quality starts to visibly deteriorate from around the fourth year onward (see Table 6), the same residents find themselves negatively affected by a decline in house prices. This lag reflects the time it takes for infrastructure disrepair

²²Table 1 reflects the relief in property tax payments for households when a road tax levy is not renewed.

to become apparent, be it through more frequent potholes or visibly eroding surfaces. By the time these problems are evident, the city's real-estate market has had sufficient time to register the disamenity, capitalizing it as a discount in property values. This dynamic captures a fundamental insight of local public finance: the preferences of current taxpayers can diverge from the longer-term public interest ([Buchanan and Tullock, 1962](#); [Alesina and Tabellini, 1990](#)).

8 Conclusion

Most studies examine the impact of new infrastructure on housing values, particularly in developing nations (Asher and Novosad, 2020; Huang, 2016; Li et al., 2016). Recent work emphasizes the challenge of endogenous placement where new projects are correlated with unobserved growth potential (Biasi, Lafortune and Schönholzer, 2025; Donaldson and Hornbeck, 2016). We circumvent this identification problem by studying the maintenance of existing infrastructure in a developed economy. This focus is particularly salient for policy-makers in the U.S. and other advanced economies, where the primary infrastructure challenge is not new construction but the preservation of a massive stock of existing assets.

By constructing a novel panel of over 3,000 renewal referendums and matching election results to local financial records and property sales, we design a quasi-experiment to estimate the causal effect of road maintenance tax cuts. We also introduce a scalable method for measuring infrastructure condition by fine-tuning an AI vision model on satellite imagery. Our results reveal that local governments failing to renew road levies face an 11% loss in maintenance funds. This fiscal shock translates into a 23% decline in road quality and, subsequently, a 9% decrease in median house prices, with effects emerging four years after the tax cut.

Our heterogeneity analysis indicates that these effects are driven by urban areas, which see a 7.6% decline in house prices, while rural areas show inconsistent effects. We also find that higher-priced homes are more sensitive to road quality shocks, and that larger tax cuts lead to larger price declines consistent with a dose-response relationship. Using a Dynamic General Equilibrium (DGE) framework, we rationalize these findings as a trade-off where the immediate benefit of lower taxes is eventually outweighed by the capitalization of public capital depreciation. Finally, our welfare analysis yields an MVPF of -1.4, implying a net welfare loss to the households due to the road maintenance tax cuts.

From a policy perspective, our findings underscore the hidden costs of deferred maintenance. While tax cuts offer short-term relief, they can trigger a vicious cycle of infrastructure decay

and asset devaluation that leaves homeowners worse off. Future research could extend our RD framework to other public goods funded by local levies, or other geographies. Advances in AI vision architectures open new avenues for researchers that could potentially allow real-time evaluation of road quality, and enable timely assessments of infrastructure policies and their welfare consequences.

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